STT 5811 HW01: Data Basics and Study Design

SOLUTIONS

Tuesday, February 10, 2026 @ 05:07 PM

Contents

Instructions

Complete exercises 2, 6, 8, 10, 12, and 18 in **IMS2E Section 1.4** and all even-numbered exercises except exercise 14 in **IMS2E Section 2.5**. Use complete sentences and explain your answers where you are asked to do so.

Section 1.4 Exercises

Exercise 2

The dataset has 19,961 observations and 9 variables.

Exercise 4

a. Based on the study findings discussed in (c), the research question would be something like: "Does explicitly telling children not to cheat affect whether or not they cheat?" Giving explicit instructions or not seems to be the explanatory variable in the experiment. Using only the description at the top of the exercise, the question would be something like: "Do characteristics like age, sex, or only-child status impact honesty or cheating." However, those are not variables that researchers can randomize in an experiment; they are more like confounding variables in this study. (Variations on this answer are possible/acceptable.)

b. The subjects are 160 children between the ages of 5 and 15.

c. There are five variables: (1) age (continuous numerical), (2) sex (categorical), (3) only child or not (categorical), (4) cheated or not (categorical), and (5) type of instructions (categorical). Note: Type of instructions is the explanatory variable; cheated or not is the response.

Exercise 6

a. The research question here is: "Is there a difference in unethical behavior between people who self-identify as having low social class versus high social class?"

b. The cases are 129 University of California at Berkeley undergraduates.

c. There are two variables: (1) self-identified low vs. high social class (categorical) and (2) number of candies the participant reported taking (discrete numerical). The scenario mentions money, education level and job respectability, but these were not variables; they were questions used to determine the social class categorization.

Exercise 8

a. The percentage of the treatment group that reported improvement is $66/85 = 0.776 = 77.6\%$.

b. The percentage of control group that reported improvement is $65/81 = 0.802 = 80.2\%$.

c. There was a (slightly) higher percentage of improvement in the control group.

d. Even if the two treatments were in fact equally effective, it is rather unlikely that we would get *exactly* the same improvement rates in the two experimental groups (samples). The observed difference of 2.6%, which equates to only about one or two participants in one group versus the other, seems like it plausibly could be due to just random chance.

e. The explanatory variable is the type of treatment the participant received (antibiotics versus placebo). The response variable is improvement in sinusitis symptoms.

Exercise 10

a. This is an experiment, because participants were randomly assigned to one of two treatment groups (vaccine or placebo) chosen by the experimenters.

b. Cases are 2,260 individuals age 12-15, with or without prior evidence of SARS-CoV-2 infection.

c. The response variable is whether or not a participant got COVID-19. It is nominal categorical.

d. The explanatory variable is whether the participant received the vaccine or placebo. They may also have used prior exposure to SARS-CoV-2 as an explanatory variable. Both of these are nominal categorical.

Exercise 12

a. Each row of the data frame represents a participant in the survey.

b. There are 1,691 survey participants.

c. The table below lists the variable types. Income is conceptually numerical, but in this survey it was recorded as an ordinal variable (as income ranges). Age is also continuous concept, but we tend to express it discretely as whole years.

variable	type
sex	nominal categorical
age	discrete numerical
marital_status	nominal categorical
gross_income	ordinal categorical
smoke	nominal categorical
amt_weekends	discrete numerical
amt_weekdays	discrete numerical

Section 2.5 Exercises

Exercise 2

The population mean is 5.5 hours and the sample mean is 6.25 hours.

Exercise 4

a. Based on the description of the wcenario, the population of interest is presumably all children age 5 and 15. The sample is 160 children in this age group.

b. It is unlikely the participants are a random (or representative) sample from the population, so the results are probably not generalizable. This study is an experiment, so if participants were randomly assigned to treatment groups (no cheating versus no instructions), the findings can be used to establish causal relationships, since randomization tends to eliminate confounding variables like age, sex, or singleton status. Refer back to Figure 2.8 in **Section 2.4**.

Exercise 6

a. The population of interest is likely people in general. However, since the sample consists of 129 UC Berkeley undergraduates, the target population is limited to UC Berkeley undergraduates.

b. If the students in this sample (who are likely not randomly sampled) can be considered representative of UC Berkeley undergraduates, then the results are generalizable to the UC Berkeley undergraduate population. If students from this school can be considered representative of all college students, or all people, we could generalize to these populations as well. However, being representative of all people is quite unlikely. The study is observational (participants were not randomly assigned to lower and upper class groups), so the findings cannot be used to establish causal relationships.

Exercise 8

a. The percentage of all videos on YouTube that are cat videos is the population parameter.

b. The value 2% is a sample statistic.

c. A video in the sample is an observation.

d. Whether or not a video is a cat video is a variable.

Exercise 10

a. This is an observational study.

b. To ensure that students from each year are fairly represented, we could use a stratified random sample. For example, randomly choose a fixed number of students from each part of campus (east and west), since different class years reside in different parts. We could also select a fixed number of students from each of the four class years, but this might be more challenging.

Exercise 12

a. This is an observational study.

b. Since the study is observational, we cannot infer causation.

c. Caffeine and lack of sleep are potential confounding variables that might explain the observed relationship between stress and muscle cramps. Muscle cramps might be the result of increased caffeine consumption and/or lack of sleep, rather directly produced by stress.

Exercise 16

a. This is an observational study.

b. The friend's statement is not justified. It implies a causal association between sleep disorders and bullying, which we cannot infer from an observational study. However, it is reasonable to say the variables are related. A better statement might be: "Children identified as bullies are more likely to be found to suffer from sleep disorders than children not identified as bullies."

Exercise 18

The estimate will be biased and tend to overestimate the true family size. Families without children have zero chance of being sampled. A family with two children is twice as likely to be sampled as a family with one child, since there are twice as many kids at the school who are available to be chosen. Even bigger families will have a proportionally larger chance.

Exercise 20

a. This is an experiment, because the researchers randomly assigned treatments to participants.

b. The explanatory variable is the Vitamin C treatment: 1g, 3g, 3g with additives, or placebo. The response variable is the duration of the cold.

c. The participants were blinded, since they did not know which treatment they received.

d. The study was double-blind with respect to the researchers who were assessing the participants, but the nurses who interacted with participants during the distribution of the medication were not blinded.

e. Participants were randomly assigned and blinded. Because of this, we would expect a similar number of people in each group to not take their pills as instructed. Non-adherence would be evenly distributed and therefore not a major confounding variable in the study.

Exercise 22

We can randomly assign participants to treatment groups: no music, instrumental, or music with lyrics. Have each person study a new concept using the same method for the same amount of time. Give them the same quiz to assess what they learned. Compare how many questions participants get correct (on average) across the three groups. (This is only one example. The important aspects are that the treatment groups are the different types of music, participants learn the same thing in the same way across groups, and they are assessed using the same method as the response variable. Random selection is not feasible in this scenario, but random assignment should be used.)

Exercise 24

a. This is an experiment.

b. The treatment group is exercise twice a week and the control group is no exercise.

c. Yes, the study uses blocking. The blocking variable is age group (18-30, 31-40, 41-55).

d. The study is not blinded. Patients definitely will know whether or not they are exercising.

e. This is an experiment, so we can make causal inferences. Because the sample is random, cause-and-effect can be generalized (cautiously, due to lack of blinding) to the population.

f. We cannot ethically require randomly selected people to participate in a study. It also might not be ethical to tell people not to exercise, or impossible due to jobs or life circumstances.

Exercise 26

a. This is a simple random sample (SRS). Usually, the SRS method is effective in getting a sample that is representative of the population. However, given the broad diversity (variability) in neighborhoods, a SRS might miss some neighborhoods that have distinct characteristics.

b. This is a stratified random sample. It would be an effective method, given the variability in neighborhoods. The sample will include representation from every neighborhood. However, it requires more work.

c. This is a cluster sample. It would *not* be an effective method, given the broad variability in neighborhoods. Clusters should be similar to one another and to the population as a whole. The sample definitely would exclude many neighborhoods with distinct characteristics, and thus not be representative.

d. This is a multi-stage sample. It starts with a cluster sample, so it has the same general flaws.

e. This is a convenience sample and not an effective method. Houses that are not close to the city council offices have zero probability of being chosen. It likely would represent only one or two neighborhoods.

Exercise 28

a. The explanatory variable is the percentage of the population with a bachelor's degree. The response variable is per capita income (in thousands). In a scatterplot, we usually think of a response variable (Y) as being a function of an explanatory variable (X).

b. There is a strong positive linear relationship between the variables. As the percent of the population with a bachelor's degree increases, the per capita income increases. There are very few counties where more than 60% of the population have a bachelor's degree and very few countries that have a more than $50,000 in per capita income (upper right of the plot).

c. This is an observational study, so we cannot make a causal statement using these data. However, we can reasonably say that having a higher percent of the population with bachelor's degree is associated with having a higher per capita income (positive association). For example, in some cases, having a higher income or coming from a family with a higher income facilitates getting a bachelor's degree, which is the reverse of the initially suggested causation.

Exercise 30

a. This is an observational study.

b. The explanatory variables are screen time, the child's sex and age, and the mother's education, ethnicity, psychological distress, and employment.

c. The response variable is psychological well-being.

d. The data come from a "nationally representative" sample, so results can be generalized to the population from which these samples were drawn.

e. Because this is an observational study, the results cannot be used to establish any causal relationships.